DATA ENGINEERING ASSIGNMENT

Assignment 1 – Apache Pig

E-Commerce Product Review Analysis – Dataset 3

Student Name	Ruchitha S
USN	4GW23CI045
Department	CSE(AI&ML)
Subject	Data Engineering
Date	24 September 2026

1. Problem Statement

An e-commerce company wants to prune invalid product reviews and find the average rating score per product category using procedural dataflow scripts. This implementation uses a new review dataset to demonstrate the Pig dataflow on different input data.

2. Objective

To use Apache Pig to load a review dataset from HDFS, remove the header, filter null ratings and ratings below 1.0, group valid reviews by product category, calculate the average rating for each category, and store the result as an HDFS output file.

3. Tools and Technologies Used

- Ubuntu Linux
- Java 17
- Apache Hadoop 3.3.6
- HDFS
- YARN / MapReduce
- Apache Pig 0.17.0
- Pig Latin

4. Dataset 3

Input file: reviews3.csv

```
/user/aiml/reviews3/reviews3.csv

review_id,product_category,rating
R201,Smartwatch,4.7
R202,Smartwatch,4.1
R203,Smartwatch,0.5
R204,Tablet,4.6
R205,Tablet,4.0
R206,Tablet,null
R207,Monitor,4.9
R208,Monitor,4.3
R209,Monitor,0.0
R210,Keyboard,3.8
R211,Keyboard,4.2
R212,Keyboard,4.6
```

5. HDFS Storage

The new dataset should be stored in a separate HDFS directory:

```
hdfs dfs -mkdir -p /user/aiml/reviews3

hdfs dfs -put ~/reviews3.csv /user/aiml/reviews3/
```

The uploaded file should be available at:

```
/user/aiml/reviews3/reviews3.csv
```

6. Pig Latin Script

```
reviews = LOAD '/user/aiml/reviews3/reviews3.csv'
          USING PigStorage(',')
          AS (review_id:chararray, product_category:chararray, rating:chararray);

data = FILTER reviews BY review_id != 'review_id';

non_null = FILTER data BY rating IS NOT NULL AND rating != 'null';

typed_data = FOREACH non_null GENERATE
             review_id,
             product_category,
             (float)rating AS rating;

valid_reviews = FILTER typed_data BY rating >= 1.0;

grouped_reviews = GROUP valid_reviews BY product_category;

average_ratings = FOREACH grouped_reviews GENERATE
                  group AS product_category,
                  AVG(valid_reviews.rating) AS average_rating;

STORE average_ratings INTO '/user/aiml/reviews3/output' USING PigStorage(',');
```

7. Explanation of Pig Operations

Operation	Purpose
LOAD	Reads the CSV dataset from HDFS.
FILTER	Removes the header, null ratings, and ratings below 1.0.
FOREACH / GENERATE	Converts rating from character data to float.
GROUP	Groups valid reviews by product category.
AVG	Calculates the average rating for each category.
STORE	Writes the final result to an HDFS output directory.

8. Execution

The Pig program should be executed in MapReduce mode using:

```
pig -x mapreduce ~/reviews_analysis3.pig
```

The output is intended to be stored at:

```
/user/aiml/reviews3/output
```

The output directory should contain:

```
_SUCCESS
part-r-00000
```

9. Output File

The expected result file is:

```
/user/aiml/reviews3/output/part-r-00000
```

It can be displayed using:

```
hdfs dfs -cat /user/aiml/reviews3/output/part-r-00000
```

10. Expected Result Calculations

Category	Valid Ratings	Expected Average
Smartwatch	4.7, 4.1	4.40
Tablet	4.6, 4.0	4.30
Monitor	4.9, 4.3	4.60
Keyboard	3.8, 4.2, 4.6	4.20

The Smartwatch rating 0.5, Tablet null rating, and Monitor rating 0.0 are excluded because the Pig script removes null values and ratings below 1.0.

11. MapReduce and YARN

The Pig program is executed in MapReduce mode. MapReduce provides the data-processing model, YARN manages application resources and execution, and HDFS provides the storage layer.

```
HDFS = Storage
MapReduce = Data Processing
YARN = Resource Management
```

12. Result Interpretation

The new dataset demonstrates that the same Pig dataflow can be reused with different review data. Invalid reviews are removed before grouping and aggregation, so the category averages are calculated only from valid ratings.

13. Conclusion

The third review dataset can be processed using Apache Pig by loading it from HDFS, removing invalid ratings, grouping valid reviews by product category, calculating average ratings, and storing the final results in HDFS.

14. Submission Checklist

- Confirm that reviews3.csv is uploaded to HDFS.
- Create and run reviews_analysis3.pig.
- Insert the actual successful terminal screenshot for Dataset 3.
- Verify the stored result using hdfs dfs -cat.
- Export the completed document to PDF if required.